

Python vs Go Parity Gaps

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Last audited: 2026-04-27 (commit 5f12be0) **Audit method:** side-by-side read of public surface (grep -E "^func" on Go, module-level inspection on Python). No behavior testing performed. **Spec reference:** docs/superpowers/specs/2026-04-26-jackett-autoconfig-clean-rebuild-design.md §7

Status definitions

- **Ported** “ feature exists in Go with equivalent behavior. Note inline divergences.
- **Partial** “ feature exists in Go but lacks a sub-behavior. Note the gap.
- **Missing** “ feature has no Go counterpart. Note user-visible risk.

Summary

Status	Count
Ported	6
Partial	4
Missing	8
Total Python features audited 18	

Matrix

download-proxy/src/merge_service/

Python module	Feature	Go location	Status	Risk
search.py	start_search() orchestrator	internal/service/merge_search.go: StartSearch	Partial	Go package “ no plugin
search.py	_classify_plugin_stderr() error categorization	(none)	Missing	No std on Go
search.py	Public tracker subprocess + NDJSON parser	(none)	Missing	Public etc.) u
search.py	Private tracker direct aiohttp (RuTracker, Kinozal, NNMClub, IPTorrents)	Partial in internal/service/merge_search.go	Partial	Verify flipping
deduplicator.py	Tiered dedup (infohash vs name+size vs fuzzy)	merge_search.go (inlined)	Partial	Verify Dedu has ex visible
enricher.py	TMDB title resolution	(none)	Missing	Search conte
enricher.py	OMDb fallback	(none)	Missing	No fa
enricher.py		(none)	Missing	

Python module	Feature	Go location	Status	Risk
	TVMaze / AniList / MusicBrainz / OpenLibrary			No m enrich
validator.py	BEP 48 HTTP scrape	(none)	Missing	No pe show
validator.py	BEP 15 UDP scrape + bencode parser	(none)	Missing	Same
scheduler.py	Recurring search driver (cron-like)	internal/api/ scheduler_api.go: ScheduleStore	Partial	Go ha drive UI let run
hooks.py	Pre/post-search + download hooks (callback dispatch)	internal/api/ hooks.go: HookStore	Partial	Same dispar
retry.py	Shared retry policy via tenacity	(none)	Missing	Each inline
jackett_autoconfig.py	Env-discovery + Jackett indexer auto-config (NEW in this spec)	(none)	Missing	Go bo â€” o

download-proxy/src/api/

Python module	Feature	Go location	Status
routes.py	merge/search REST endpoints	internal/api/ search.go	Ported
auth.py	Private-tracker auth (login, session refresh, CAPTCHA)	(none)	Missing
hooks.py	hooks REST endpoints	internal/api/ hooks.go	Ported
scheduler.py	scheduler REST endpoints	internal/api/ scheduler_api.go	Ported (storage only â€” see scheduler driver gap above)
streaming.py	SSE streaming endpoint	internal/ service/ sse_broker.go + internal/api/ search.go	Ported
theme_state.py	Dashboard theme state	internal/api/ theme.go	Ported
jackett.py	/api/v1/ jackett/ autoconfig/last (NEW)	(none)	Missing

download-proxy/src/ui/

Python module	Feature	Go location	Status
Jinja2 dashboard templates	Operator UI fallback	(none)	Missing

The Angular SPA dashboard is shared (lives in frontend/, served by both). The Jinja2 fallback dashboard is Python-only.

User-visible regressions if you flip Go-only today

In priority order:

1. **No metadata enrichment.** Search results lose posters, year, content type. Highest user-visible degradation.
2. **No Jackett auto-config.** Every fresh boot requires manual Jackett UI clicks.
3. **No public tracker plugin fan-out.** 600+ Python plugins (eztv, piratebay, etc.) unreachable. Search would only return whatever Jackett aggregates + the 4 private trackers (and only those wired in Go).
4. **No tracker validation.** Stale peer counts, no health detection.
5. **Schedules silently no-op.** UI creates them, driver never fires.
6. **No private-tracker auth REST surface.** No CAPTCHA refresh flow.

Per-gap follow-up specs (proposed, in priority order)

1. `enricher.go` port “biggest user-visible gap. Wraps 6 external APIs. Estimate 2-3 days with tests.
2. `validator.go` port “BEP 48/15 with bencode parser. Estimate 1-2 days.
3. `scheduler.go` **driver loop** “make existing storage actually fire schedules. Estimate 1 day.
4. **Public tracker plugin subprocess fan-out** “replicate `_search_tracker()` subprocess pattern. Estimate 2-3 days.
5. `jackett_autoconfig.go` port of `jackett_autoconfig.py`. Estimate 1 day.
6. `auth.go` REST surface for private-tracker auth flows. Estimate 1-2 days.
7. `hooks.go` **runtime dispatch** “verify and complete callback firing.
8. `retry.go` shared policy.
9. `deduplicator.go` explicit tiered structure (clarification, not new behavior).

Each gap should ship as its own spec + plan + implementation cycle in the superpowers flow, so nothing gets lost during the eventual Go-only flip.

Audit data points captured

- Go service exports (count): 18 functions across `merge_search.go` + `sse_broker.go`.
- Go API exports (count): ~30 handler/store functions across 6 files.
- Python `merge_service/` modules: 7 (deduplicator, enricher, hooks, jackett_autoconfig, retry, scheduler, search, validator).
- Python `api/` modules: 7 (auth, hooks, jackett, routes, scheduler, streaming, theme_state).